

CODE:

```
def getAction(state):
    Curr_State=state
    if Curr_State[0]=='R' and Curr_State[1][1]=='D':
        return "SUCK"
    elif Curr_State[0]=='L' and Curr_State[1][0]=='D':
        return "SUCK"
    elif Curr_State[0]=='R' and Curr_State[1][1]=='C':
        return "MOVELEFT"
    elif Curr_State[0]=='L' and Curr_State[1][0]=='C':
        return "MOVERIGHT"
    else:
        return None

def updateState(state, action):
    Curr_State=state
    if Curr_State[0]=='R' and action=="SUCK":
        Curr_State[1][1]='C'
    elif Curr_State[0]=='L' and action=="SUCK":
        Curr_State[1][0]='C'
    elif Curr_State[0]=='R' and action=="MOVELEFT":
        Curr_State[0]='L'
    elif Curr_State[0]=='L' and action=="MOVERIGHT":
        Curr_State[0]='R'
    return Curr_State

def bool_all_rooms_clean(state):
    if state[1][0]=='C' and state[1][1]=='C':
        return True
    else:
        return False

def run_vacuum(start_state):
    state = start_state
    loop_count = 0
    while not bool_all_rooms_clean(state):
        action = getAction(state)
        state = updateState(state, action)

    print(f"cycle:{loop_count}\taction={action}\t\tstate={state}")
    loop_count += 1
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        if loop_count > 20:
            print(
                "Uh oh ! Code stuck in a state ... program terminated")
            return

    print("All rooms clean!")
    return

def main():
    start_state = ['R', ['D', 'D']]
    print(f"\nPart1. Start in room R\nStart State: {start_state}")
    run_vacuum(start_state)

    start_state = ['L', ['D', 'D']]
    print(f"\nPart2. Start in room L\nStart State: {start_state}")
    run_vacuum(start_state)

if __name__ == '__main__':
    main()

```

OUTPUT:

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Start State: ['R', ['D', 'D']]
cycle:0      action=SUCK      state=['R', ['D', 'C']]
cycle:1      action=MOVELEFT  state=['L', ['D', 'C']]
cycle:2      action=SUCK      state=['L', ['C', 'C']]
All rooms clean!

```